=== UNIT 3: FIBER OPTICS & STRUCTURAL PHOTONIC SENSING ===

>> 3.1 Total Internal Reflection and Optical Waveguiding

An optical fiber is a cylindrical dielectric waveguide made of high-purity silica glass (SiO2) that transmits optical signals via repeated Total Internal Reflection (TIR). In civil engineering, optical fibers form telecommunication networks and serve as distributed sensors embedded in concrete dams, bridge cables, and tunnels for Structural Health Monitoring (SHM).
* Anatomy of an Optical Fiber: (1) Core: Central dielectric cylinder of high refractive index n1. (2) Cladding: Surrounding coaxial dielectric sheath of slightly lower refractive index n2 (n1 > n2). (3) Primary Buffer Jacket: Protective polymeric acrylate coating that absorbs mechanical abrasion and moisture.
* Critical Angle (theta_c): By Snell's Law at core-cladding boundary: n1 * sin(theta_c) = n2 * sin(90) => sin(theta_c) = n2 / n1.
Total internal reflection occurs only when internal ray angle of incidence phi exceeds theta_c: phi > theta_c.

>> 3.2 Acceptance Angle, Numerical Aperture, and Index Contrast

* Acceptance Angle (theta_a): The maximum half-angle of the external acceptance cone at the fiber launch face within which launched rays undergo TIR down the core:
Applying Snell's law at air-core entry interface: n_0 * sin(theta_a) = n1 * sin(90 - theta_c) = n1 * cos(theta_c) = n1 * sqrt(1 - sin^2(theta_c)) = n1 * sqrt(1 - (n2/n1)^2) = sqrt(n1^2 - n2^2).
Assuming launch in air (n_0 = 1.0): sin(theta_a) = sqrt(n1^2 - n2^2). Acceptance angle: theta_a = asin(sqrt(n1^2 - n2^2)).
* Numerical Aperture (NA): A dimensionless figure of merit quantifying the light-gathering capacity of the fiber:
Formula: NA = sin(theta_a) = sqrt(n1^2 - n2^2).
* Fractional Refractive Index Contrast (Delta): Relative refractive index difference:
Delta = (n1 - n2) / n1 ~= (n1^2 - n2^2) / (2 * n1^2).
Approximation for weakly guiding fibers (Delta << 1): NA = n1 * sqrt(2 * Delta).

>> 3.3 Classification of Optical Fibers

* 1. Step-Index Single-Mode Fiber (SMF): Extremely narrow core diameter (2*a ~ 8 to 10 micrometers) with uniform index n1 and sudden drop to n2. Only the fundamental spatial mode (HE11) propagates. Exhibits zero intermodal dispersion, enabling ultra-long-haul telecommunications (terabit/s over 100+ km).
* 2. Step-Index Multi-Mode Fiber (MMF): Large core diameter (50 to 100 micrometers). Supports hundreds of spatial modes. Susceptible to significant intermodal dispersion,

limiting reach to local networks (< 2 km).
* 3. Graded-Index Multi-Mode Fiber (GRIN): Core refractive index decreases continuously as a parabolic profile from center n1 to cladding boundary n2: n(r) = n1 * sqrt(1 - 2*Delta*(r/a)^alpha). Off-axis rays travel through lower-index outer regions at higher phase velocity, equalizing propagation transit times across modes and dramatically reducing modal dispersion.

>> 3.4 Normalized Frequency (V-Number) and Mode Count

* V-Number (Normalized Frequency): Dimensionless parameter determining optical propagation characteristics:
Formula: V = (2 * pi * a / lambda) * NA = (2 * pi * a / lambda) * sqrt(n1^2 - n2^2), where a is core radius.
* Single-Mode Cutoff Condition: A step-index fiber operates strictly in single mode if: V <= 2.4048 (the first root of Bessel function J0). Cutoff wavelength: lambda_c = (2 * pi * a * NA) / 2.4048.
* Total Bound Modes in Multimode Fibers:
- Step-Index MMF: N_modes ~= V^2 / 2.
- Graded-Index MMF (parabolic alpha = 2): N_modes ~= V^2 / 4.

>> 3.5 Attenuation Mechanisms in Optical Fibers

Signal loss (attenuation) in dB/km is defined by: alpha_dB = (10 / L) * log10(P_in / P_out):
* 1. Material Absorption: (a) Intrinsic absorption from electronic band transitions in UV and molecular vibrations of Si-O bonds in far-IR. (b) Extrinsic absorption from trace transition metal ions and hydroxyl (OH-) radical impurity overtones (causing absorption water peaks at 950 nm, 1240 nm, and 1380 nm).
* 2. Rayleigh Scattering Loss: Microscopic density and refractive index fluctuations frozen into the glass during cooling. Strongly wavelength-dependent: alpha_Rayleigh proportional to 1 / lambda^4. Dominates at short wavelengths, dictating the selection of infrared transmission windows (1310 nm and 1550 nm).
* 3. Bending Losses: (a) Macrobending: Loss occurring when fiber bend radius falls below critical curvature (~few cm), causing core rays to strike cladding at angles less than critical angle theta_c. (b) Microbending: Microscopic axial random deflections caused by irregular mechanical pressure or buffer jacket thermal shrinkage.

>> 3.6 Pulse Dispersion and Bandwidth Limitations

* Dispersion: Temporal pulse spreading as optical pulses propagate along fiber length, causing inter-symbol interference (ISI) and limiting maximum bit rate B:
* Intermodal Dispersion (Modal): Occurs only in multimode fibers because higher-order

zigzag modes travel longer physical optical path lengths than axial modes: Delta t_modal = (L * n1 / c) * Delta.
* Intramodal (Chromatic) Dispersion: Occurs in all fibers due to finite spectral width of light source:
- Material Dispersion: Phase velocity varies with wavelength because refractive index n(lambda) is wavelength-dependent.
- Waveguide Dispersion: Distribution of optical power between core and cladding varies with wavelength.

>> 3.7 Fiber Bragg Grating (FBG) Sensors in Civil Engineering

A Fiber Bragg Grating is a periodic modulation of core refractive index formed by UV interference lithography:
* Bragg Reflection Condition: Reflects a narrow wavelength band centered at Bragg wavelength lambda_B: lambda_B = 2 * n_eff * Lambda, where Lambda is grating pitch.
* Sensing Principle: External axial mechanical strain (epsilon) and thermal changes (Delta T) physically alter grating period Lambda and effective index n_eff: Delta lambda_B / lambda_B = (1 - p_e) * epsilon + (alpha_th + xi) * Delta T.
* Applications: Embedding arrays of multiplexed FBGs inside prestressed concrete girders, suspension bridge stay-cables, and tunnel linings to record real-time internal micro-strain, temperature variations, and crack inception without electromagnetic interference.

[EXAMPLE 1: Acceptance Angle and Numerical Aperture of Silica Fiber]

Problem Context: A step-index silica optical fiber has a core refractive index n1 = 1.480 and a cladding refractive index n2 = 1.455. The fiber launch takes place in air (n_0 = 1.000). Calculate: (1) Critical angle theta_c at core-cladding interface, (2) Numerical Aperture NA, (3) Acceptance angle theta_a in air, (4) Fractional index difference Delta.
Solution Step 1: Compute Critical Angle theta_c:
sin(theta_c) = n2 / n1 = 1.455 / 1.480 = 0.98311.
theta_c = asin(0.98311) = 79.45 deg.
Solution Step 2: Compute Numerical Aperture NA:
NA = sqrt(n1^2 - n2^2) = sqrt((1.480)^2 - (1.455)^2) = sqrt(2.1904 - 2.117025) = sqrt(0.073375) = 0.2709.
Solution Step 3: Compute Acceptance Angle theta_a in air:
sin(theta_a) = NA = 0.2709 => theta_a = asin(0.2709) = 15.72 deg.
Total acceptance cone angle = 2 * theta_a = 31.44 deg.
Solution Step 4: Compute Fractional Index Difference Delta:
Delta = (n1 - n2) / n1 = (1.480 - 1.455) / 1.480 = 0.025 / 1.480 = 0.01689 (1.69%).
Conclusion: theta_c = 79.45 deg; NA = 0.2709; theta_a = 15.72 deg; Delta = 1.69%.

[EXAMPLE 2: Single-Mode Core Radius Design for Telecom Window]

Problem Context: An optical telecom link operates at laser wavelength lambda = 1.31 micrometers (1310 nm). The core index is n1 = 1.465 and cladding index is n2 = 1.460. Determine: (1) Maximum allowable core radius a_max for single-mode operation, (2) If core diameter is fabricated at 2*a = 8.0 micrometers, verify single-mode propagation and calculate cutoff wavelength lambda_c.

Solution Step 1: Compute NA:

NA = sqrt(1.465^2 - 1.460^2) = sqrt(2.146225 - 2.1316) = sqrt(0.014625) = 0.1209.

Solution Step 2: Single-Mode Condition (V <= 2.4048):

V = (2 * pi * a / lambda) * NA <= 2.4048.

a_max = (2.4048 * lambda) / (2 * pi * NA) = (2.4048 * 1.31 * 10^-6) / (2 * pi * 0.1209) = (3.1503 * 10^-6) / 0.7596 = 4.147 * 10^-6 m = 4.15 micrometers.

Maximum core diameter = 2 * a_max = 8.30 micrometers.

Solution Step 3: Verify operation for 2*a = 8.0 micrometers (a = 4.0 micrometers):

V = (2 * pi * 4.0 * 10^-6 / 1.31 * 10^-6) * 0.1209 = (19.185) * 0.1209 = 2.320 < 2.4048 (Single mode confirmed!).

Cutoff wavelength: lambda_c = (2 * pi * a * NA) / 2.4048 = (2 * pi * 4.0 * 10^-6 * 0.1209) / 2.4048 = 1.264 micrometers (1264 nm).

Conclusion: Maximum core radius a_max = 4.15 micrometers; At 2*a = 8 micrometers, V = 2.320 and cutoff wavelength lambda_c = 1264 nm.

[EXAMPLE 3: Optical Link Attenuation and Repeater Spacing]

Problem Context: A civil engineering fiber optic backbone connecting sensor telemetry along a 40 km highway corridor uses single-mode fiber with attenuation coefficient alpha = 0.22 dB/km at lambda = 1550 nm. The link has 8 fusion splices (loss = 0.10 dB per splice) and two connector pairs at endpoints (loss = 0.75 dB per connector pair). Input optical power is P_in = 5.0 mW (+7 dBm). Receiver sensitivity requires minimum P_min = 0.05 microwatts (-43 dBm). Determine: (1) Total link loss in dB, (2) Output power P_out, (3) System safety margin.

Solution Step 1: Calculate Total Link Attenuation Loss:

Fiber cable loss = alpha * L = 0.22 dB/km * 40 km = 8.80 dB.

Splice losses = 8 * 0.10 dB = 0.80 dB.

Connector losses = 2 * 0.75 dB = 1.50 dB.

Total Link Loss alpha_total = 8.80 + 0.80 + 1.50 = 11.10 dB.

Solution Step 2: Compute Output Power P_out:

Total loss in dB = 10 * log10(P_in / P_out) => 11.10 = 10 * log10(5.0 mW / P_out).

log10(5.0 mW / P_out) = 1.110 => 5.0 mW / P_out = 10^(1.110) = 12.8825.

P_out = 5.0 mW / 12.8825 = 0.3881 mW = 388.1 microwatts.

Solution Step 3: Compute Power Margin above Receiver Sensitivity:

In dBm: P_out(dBm) = 10 * log10(0.3881 mW / 1 mW) = - 4.11 dBm.

Receiver sensitivity P_min = - 43.0 dBm.
Safety Margin = P_out(dBm) - P_min(dBm) = - 4.11 - (- 43.0) = 38.89 dB.
Conclusion: Total link loss = 11.10 dB; Received power = 388.1 microwatts; Generous safety margin of 38.89 dB.

[EXAMPLE 4: Intermodal Dispersion and Maximum Transmission Bit Rate]

Problem Context: An industrial step-index multimode fiber has core index n1 = 1.500, length L = 3.0 km, and fractional index contrast Delta = 0.010 (1.0%). Calculate: (1) Minimum and maximum transit times of light rays through the fiber, (2) Total intermodal pulse broadening Delta t_modal, (3) Maximum theoretical bit rate B_max without inter-symbol interference.
Solution Step 1: Transit Times for Axial and Highest-Order Rays:
Speed of light in core: v = c / n1 = (3.0 * 10^8 m/s) / 1.500 = 2.0 * 10^8 m/s.
Minimum transit time (axial direct ray): t_min = L / v = (L * n1) / c = (3000 m * 1.50) / (3.0 * 10^8 m/s) = 1.50 * 10^-5 s = 15.0 microseconds.
Maximum transit time (steepest zigzag ray at critical angle theta_c): t_max = (L * n1^2) / (c * n2) ~= t_min / (1 - Delta).
t_max = 1.50 * 10^-5 / (1 - 0.010) = 1.50 * 10^-5 / 0.99 = 1.51515 * 10^-5 s = 15.1515 microseconds.
Solution Step 2: Pulse Broadening Delta t_modal:
Delta t_modal = t_max - t_min = (L * n1 / c) * Delta = (1.50 * 10^-5 s) * 0.010 = 1.50 * 10^-7 s = 150 nanoseconds.
Solution Step 3: Maximum Bit Rate B_max:
To prevent pulses from overlapping, bit duration T_bit >= 2 * Delta t_modal:
B_max ~= 1 / (2 * Delta t_modal) = 1 / (2 * 150 * 10^-9 s) = 1 / (3.0 * 10^-7 s) = 3.33 * 10^6 bits/s = 3.33 Mbps.
Conclusion: Intermodal delay spread Delta t = 150 ns; Maximum data rate is limited to 3.33 Mbps over 3 km.

>> 3.8 University Review & Examination Questions

* Q1: Derive expressions for Numerical Aperture and Acceptance Angle of a step-index optical fiber.
Answer: Consider a cylindrical fiber with core index n1 and cladding n2 (n1 > n2) in an ambient medium of index n0. A ray enters the core face at angle theta_0 and refracts at angle r. At the core-cladding interface, the angle of incidence is phi = 90 - r. For total internal reflection to occur, phi >= theta_c, where sin(theta_c) = n2 / n1. At the critical limit phi = theta_c, cos(r) = sin(phi) = n2 / n1. By Snell’s law at entrance:
n0 * sin(theta_a) = n1 * sin(r) = n1 * sqrt(1 - cos^2(r)) = n1 * sqrt(1 - (n2/n1)^2) = sqrt(n1^2 - n2^2). For air launch (n0 = 1.0): sin(theta_a) = sqrt(n1^2 - n2^2).
Numerical Aperture is defined as NA = sin(theta_a) = sqrt(n1^2 - n2^2). The acceptance angle is theta_a = asin(sqrt(n1^2 - n2^2)).

* Q2: Explain the working principle of Fiber Bragg Grating (FBG) sensors and their direct engineering utility in Civil Structural Health Monitoring.
Answer: An FBG sensor is created by exposing the germanium-doped silica core of a single-mode fiber to a periodic ultraviolet laser interference pattern, inducing a permanent periodic modulation of refractive index of spatial pitch Lambda. When broadband light travels down the fiber, a narrow spectral peak is reflected at the Bragg wavelength: lambda_B = 2 * n_eff * Lambda. When the host civil structure undergoes mechanical strain or temperature fluctuations, the physical pitch Lambda stretches or compresses, shifting the reflected Bragg wavelength by Delta lambda_B. In civil infrastructure (bridges, dams, wind turbine blades), multiplexed FBG arrays are cast directly into reinforced concrete to monitor strain distribution, crack opening, and thermal gradients continuously over decades without electrical noise or corrosion.

>> 3.9 Key Takeaways and Revision Points

* Total internal reflection requires n1 > n2 and angle of incidence phi > theta_c.
* Numerical Aperture NA = sqrt(n1^2 - n2^2) measures light-gathering capability.
* Single-mode fiber cutoff requires normalized frequency V <= 2.4048.
* Intermodal dispersion is eliminated in single-mode fibers; FBG sensors measure strain via lambda_B shift.